

JMO – 2009

Max Mark – 100

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Q 1 to 11 carry 8 marks each and Q 12 carries 12 marks.
- Answer all the questions.

- How many zeroes are there at the end of product of 20^{50} and 50^{20} .
 - The time on a digital clock is 5:55. How many minutes will pass before the clock next shows a time with all digits identical?
- In an election, Jatin received 60% of the votes and Samir received all the rest. If Jatin won by 24 votes, how many people voted?
 - When a positive integer N divided by 60, the remainder is 49. What is the remainder when N is divided by 15?
- What is the sum of first 2005 terms of the sequence 1, 2, 3, 4, 1, 2, 3, 4, 1, 2, 3, 4,
 - Seven children, each with same birth day, were born in seven consecutive years. The sum of ages of youngest three children is 42. What is the sum of the ages of the oldest three?
- If x and y are each chosen from the set of integers {1, 2, 3, 5, 10} then what is the largest possible values of $\frac{x}{y} + \frac{y}{x}$?
 - When 14 is divided by 5 remainder is 4. When 14 is divided by a positive integer n the remainder is 2. For how many different values of n is this possible?
- Find the sum of all the digits of the result of the subtraction $10^{99} - 99$.
 - Find the largest among the numbers: 2222, 222^2 , 22^{22} , 2^{222} .
- Smruti leaves school at the same time every day. If she cycles at 20 km/h she arrives home at 4.30 PM. If she cycles at 10 km/h she arrives at 5.15 PM. At what speed in km/h must she cycle to arrive home at 5 PM?
- ABC is a triangle with the length of sides as integer in the ratio 3:4:5.
 - If one of the side of ABC is 28, then what are other sides?
 - Find the length of the sides of ABC which has perimeter 96.
- The set {1, 4, n} has property that when any two distinct elements are chosen and 2112 is added to their product, the result is a perfect square. If n is a positive integer what is the number of possible values of n?
- If $a > 0$ and $b > 0$ an operation * is defined as $a * b = \frac{a+b}{1+ab}$ then
 - Calculate $2 * 5$
 - If $2 * x = \frac{5}{7}$, then find the value of x.

- (iii) If for some values of x and y , $x * y = \frac{x+y}{12}$, then determine all possible integer pairs (x, y) .
10. The sequence 2, 3, 5, 6, 7, 10,..... consists of all positive integers which are not perfect powers. What is the sum of squares of the digits of 1000th number in the sequence? (where $25 = 5^2, 16 = 2^4, 8 = 2^3$ are perfect powers)
11. In the sum $2 + 22 + 222 + 2222 + \dots$ up to 101 terms, determine the digits in 1's place, 10's place, 100's place and in the middle position.
- 12.

- (a) A Class of 20 students was given a two question quiz. The results are as listed below:

Question No.	No. of students answered correctly
1	18
2	14

Determine the smallest possible number and the largest possible number of students that could have answered all two questions correctly. Explain why these are smallest and largest possible numbers.

- (b) A Class of 20 students was given a two question quiz. The results are as listed below:

Question No.	No. of students answered correctly
1	18
2	14
3	12

Determine the smallest possible number and the largest possible number of students that could have answered all three questions correctly. Explain why these are smallest and largest possible numbers.

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